Dear Referee,

Thank you for carefully reading the previous version of the paper and providing me with great comments and suggestions. I think that the paper has improved substantially thanks to your suggested revisions.

This letter has two sections:

1. The first section summarizes the key changes to the manuscript.
2. The second section reports a detailed answer to each of the comments and suggestions in your letter. For each point, I first report your comment (in blue) and then discuss the paper modifications and extended analysis that address it.

Sincerely,

Lulu Wang

Summary of the Main Changes

This revision strengthens the reduced-form evidence, clarifies data limitations and modeling assumptions, estimates alternative specifications with debit-credit substitution, expands the counterfactual analysis, and rewrites the text for a broader audience.

1. **Reduced Form.** The Durbin section now emphasizes that interchange fees steer payment choices through multiple channels (rewards, advertising, teller incentives). New bank-level data show that among the banks that paid debit rewards pre-Durbin, those that ended rewards saw debit card volumes grow more slowly than those that kept rewards by a margin similar to the baseline difference-in-difference estimate. I also clarify the selectedness of the merchant event studies and expand the Discover analysis with a formal fixed-effects specification.

2. **Modeling.** A new "Key Assumptions" discussion clarifies what the data can and cannot identify. Merchant-type estimation is framed as a calibration anchored by the grocery-chain event studies. Fixed costs of acceptance cannot be separately identified. The counterfactuals are short-run predictions that hold non-rewards characteristics fixed.

3. **Credit-Debit Substitution.** The baseline keeps credit and debit segmented at the point of sale. Antitrust testimony and consumer surveys support segmentation, but Discover's rewards experiment shows some substitution at the margin. Appendix OF estimates two specifications that relax segmentation; welfare results are close to the baseline for every counterfactual other than uncapping debit.

4. **Counterfactuals.** The main fee cap is now 120 bps, half-way between the previous cap and current levels. The relationship between cap strictness and welfare is concave, so this cap captures most of the welfare gains. New analyses decompose welfare gains into channels and model a dual-routing mandate. Re-estimating with income-dependent credit access yields similar results ($31 bn vs. $27 bn).

5. **Exposition.** The introduction is rewritten for a broader audience, built around standard IO concepts with the Durbin Amendment introduced in the opening paragraphs. Only three appendices are primary, with supplementary material in a separate Online Appendix. Footnotes and institutional digressions are shortened throughout.

Detailed Response to Referee 1

I organize my responses as follows. I begin with the four comments assessed as "no response" or "unsatisfactory" in the first round. I then address new comments from the current round, and close with follow-up clarifications on comments assessed as mostly or somewhat satisfactory. Comments assessed as fully satisfactory are not repeated here.

Previously Unaddressed and Unsatisfactory Comments

> The assumption that a merchant enjoys a fixed boost in sales γ when selling to a consumer who uses a card has some undesirable properties. When consumers carry both a credit and a debit card (as I suspect most do in practice), then the merchant has no incentive to accept credit cards; credit cards offer no incremental sales increase over debit cards.
>
> No response: I could not locate a response to this comment in the article. I struggle to imagine that consumers could carry two credit cards without carrying a debit card, which I had thought necessarily follows from having a bank account. If credit card consumers also have a debit card, and incremental consumer spending is constant across debit and credit cards, the fact that merchants adopt high-free credit cards seems to be an artifact of the somewhat arbitrary assumption that consumers may carry only up to two cards. (I may be missing something here.)

Reply: You are correct that γ is the same for credit and debit transactions in the baseline model. Factually, as shown in Table 1, most credit card holders also own a debit card.

The reason merchants still have an incentive to accept credit in the model is that the baseline assumes credit and debit serve different transaction types (the segmentation assumption discussed in Section IV.F). Under segmentation, a consumer who prefers credit for a given transaction pays with cash, not debit, if credit is unavailable, so a debit card in the same wallet does not blunt merchants' incentive to accept credit.

The two-card representation is consistent with observed behavior — around 95% of card spending is concentrated on just two networks (Online Appendix Table OA.4) — but the segmentation assumption is doing the substantive work. The concentration is also informative about preferences: a consumer placing 95% of her spending on two networks is revealing that the third-best card is a poor substitute, not an equally attractive option that the model arbitrarily excludes.

The reduced-form evidence in Section III.B supports the idea that accepting credit cards in addition to debit cards increases sales. When a grocery retailer begins accepting credit cards, whether a consumer used credit at *other* stores is strongly predictive of the sales response at the newly accepting merchant. This indicates that card ownership alone is not sufficient and that usage patterns matter, but it does not rule out some degree of substitution.

The assumption that consumers do not substitute between credit and debit at the point of sale may seem strong, but antitrust testimony in *Ohio v. AmEx* and the Durbin Amendment's failure to move credit interchange both support it. Online Appendix OF estimates two alternatives that relax it, one allowing substitution with incremental debit sales relative to cash and another treating debit as generating no incremental sales. These alternatives generate counterfactual predictions for the drivers of merchants' credit card acceptance decisions. Nonetheless, welfare results are close to the baseline for every counterfactual other than uncapping debit fees, where the substitution channel by construction matters most.

> Relatedly, the modelling of rewards is unnatural. Consumers enjoy a boost to their income for adopting a card even if they never use the card.
>
> I think that the author may have tried to respond to this comment with the text on page 21 reading "The pecuniary utility for single- homing agents is micro-founded in the consumer's optimal consumption problem across merchants. The optimized value of log utility for a consumer with CES preferences that generate the demand curve in Equation 28 is approximately log(U)." This requires more explanation; the microfoundation is not explicitly provided and is not clear to me.

Reply: In the model, consumers always use their preferred payment method at merchants that accept it. There is no choice about whether to use the card conditional on adoption. The concern about consumers receiving rewards without using the card does not arise in equilibrium: merchants accept cards, and consumers always use their preferred payment method at accepting merchants.

From the consumer's perspective, the only difference between a per-transaction reward and a lump-sum reward of equal monetary value is that a lump-sum reward has no effect on the allocation of spending across merchants. This is consistent with the Discover reduced-form evidence (Online Appendix OD.2). When Discover offers higher rewards, consumers switch payment methods but do not reallocate shopping trips across stores.

Section IV.B.2 now presents the consumer's CES utility function (Equation 4) and derives the demand function (Equation 5) and pecuniary utility (Equation 16) directly. Rewards f^w enter as a lump-sum income boost: total expenditure is $y(1+f^w)$, and log indirect utility reduces to $\log U^w = \log y + f^w - \log P^w$. By Roy's identity, whether f^w is paid out lump sum or on each transaction has no effect on the optimized value of indirect utility.

"Another concern relates to merchant pricing. In general, markups and pass-through depend on the slope and curvature of demand, respectively. CES demand has one parameter that governs both markups and pass-through."

Mostly unsatisfactory response: in Section IV.F.2, the author claims that "Appendix F.6 extends the model to allow for an arbitrary degree of pass- through." This is not true. Appendix F.6 provides analysis of a case with no pass-through. I did not see anything in Appendix F.6 about allowing an arbitrary degree of pass-through; the misleading statement in the main text is the main reason why I have labelled the response as "mostly unsatisfactory." With that said, the additions in Appendix F.6 with no pass-through are much appreciated and interesting. I would suggest that the author acknowledges the problems associated with full pass-through and explains what happens in the opposite case with no pass-through. Upon making these changes and removing the false statement from the text, I would update my assessment to "Fully satisfactory response."

Reply: I have clarified in the main text that Online Appendix OG.1 considers the polar opposite case of no pass-through, as you suggested. I also discuss why data limitations prevent me from directly estimating pass-through.

"Last, there is a concern in the "increased competition" counterfactual that adding a new credit card network mechanically increases credit card usage by giving consumers new logit shocks for credit cards. Given that credit card usage is welfare reducing in the model, adding a new card is thus welfare reducing before the pro-competitive effects of entry (changes in prices/fees) are considered. It would be good to take note of this concern, even if it is not addressed via a change in the counterfactual."

No response: I could not find a response to this comment in the article. Although the "increased competition" counterfactual has been removed in favour of a focus on monopolization, my original comment also applies to a comparison of a monopoly regime with the baseline regime.

Reply: Your original concern about the entry counterfactual was valid: adding a new credit card network introduces new logit shocks that mechanically increase credit card adoption. When credit cards are welfare-reducing, this mechanical effect creates a bias against entry in the counterfactual.

However, the concern does not apply to the monopoly counterfactual. Monopolization changes ownership (one firm sets fees for all networks to maximize joint profits) but does not change the set of available products nor the realized logit shocks. Any changes in adoption patterns are driven entirely by changes in equilibrium fees and rewards, not by mechanical changes to preferences.

New Comments from Current Round

> It would be interesting to decompose the channels by which the fee cap affects welfare by computing the counterfactual (i) holding fixed prices and merchant adoption, (ii) holding fixed prices, (iii) holding fixed merchant adoption, and (iv) holding fixed neither (which, of course, is what the author is currently doing). I suppose that merchants compete away most of their gains from fee caps by reducing prices following the fee cap. Given the multi-sided nature of the market, it would also be interesting to see how participation changes on the merchant side affect consumer welfare and whether this dynamic is relevant at all for the welfare results.

Reply: I now include a full decomposition in Table 6, which sequentially allows merchant responses to fee caps (Section VI.A.2).

Your intuition is exactly right: merchants compete away nearly all of their direct gains from lower fees. With prices and acceptance fixed (first row), merchants pocket \$50Bn in fee savings, but once they re-optimize prices (second row), competition dissipates \$49Bn of those savings into lower consumer prices.

The more surprising finding is *why* consumers gain \$28Bn overall. Fees and rewards fall by roughly equal amounts, so the direct transfer to consumers is small. The welfare gains come from reduced credit card adoption. Under price coherence, each credit card user imposes an externality through higher retail prices, so the marginal consumer's rewards are a cross-subsidy funded by all shoppers. Fee caps shrink this cross-subsidy, eliminating real costs such as debt aversion, that rewards were compensating for.

Total welfare in the first row rises by only \$14Bn because lower rewards contract consumption when retail prices are fixed. Since margins are positive, reduced consumption creates deadweight loss. Merchant price cuts in the second row reverse this contraction, so the full benefit of reduced credit aversion materializes only across both rows together. The third row (Merchant Adoption) has a negligible net welfare effect.

> The results regarding the effects of competition on platform fees in Teh et al. suggest that multihoming plays a crucial role in determining whether competition tends to raise or lower merchant fees. To my understanding, there is no empirical paper that estimates how multihoming affects the extent to which competition shifts the relative balance of consumer fees (or rewards) and merchant fees. The author has an opening to do exactly this by running the monopoly counterfactual under alternative assumptions regarding consumer multihoming. This exercise would also suggest how monopolization would affect welfare in payment card (or, more generally, platform markets) with different levels of multihoming than the US payment card market.

Reply: I have added a "Dual Routing" counterfactual that increases the share of credit card holders who multihome from approximately 60% to approximately 80%.

Tables OA.12 and 5 show the results. Consistent with the theoretical predictions of Teh et al. (2023), increased multihoming leads networks to compete more intensely for transactions rather than for exclusive relationships. Credit card fees decline by 13 bp (SE 3.2) and rewards by 15 bp (SE 3.4), because networks have less incentive to offer generous rewards when targeted cardholders already hold competing cards. Despite lower rewards, consumer welfare still increases by $3.3 billion (SE 1.4), net of the mechanical χ^{22} adjustment described in Online Appendix OE.7.

The exercise varies multihoming under the competitive market structure rather than under monopoly. The Credit Card Competition Act is the closest policy analogue and would increase multihoming without changing market structure.

> How was the large retailer in Section III.B chosen? Are there any other retailers in the sample period that began accepting credit cards, to provide a sense of effects beyond a single retailer and beyond the grocery category?

Reply: At the time of the event studies, nearly all large retailers already accepted credit cards, so the grocers I study are highly selected. The event study is not meant to be representative of the average retailer but rather the marginal merchant.

To identify stores, I screened store-level changes in the share of payments on credit cards over time and verified that these shifts correspond to changes in credit card acceptance reported in news articles and press releases. I excluded events involving warehouse stores because those typically involve changes in *which* credit cards are accepted (accompanied by coordinated card-switching campaigns) rather than *whether* credit cards are accepted, which would confound the effect I am trying to measure. The Kilts data use agreement requires retailer anonymity. Only two events in the sample period meet these criteria, which is a limitation of the design.

> The article uses the number of trips as the outcome when studying merchant benefits from card acceptance, claiming that revenue is unsuitable for use due to its fat tails. But the retailer considers sales effects, not foot traffic effects, when choosing whether to adopt a payment card. Wouldn't a log transform of revenue deal with the fat tails?

Reply: The main text regression now uses dollar purchases at the consumer level as the main outcome to capture both trips and spending per trip.

Why use a relatively exotic Poisson model as opposed to a more standard linear model in the analysis of merchant benefits from card acceptance? Are the results from a linear model qualitatively similar?

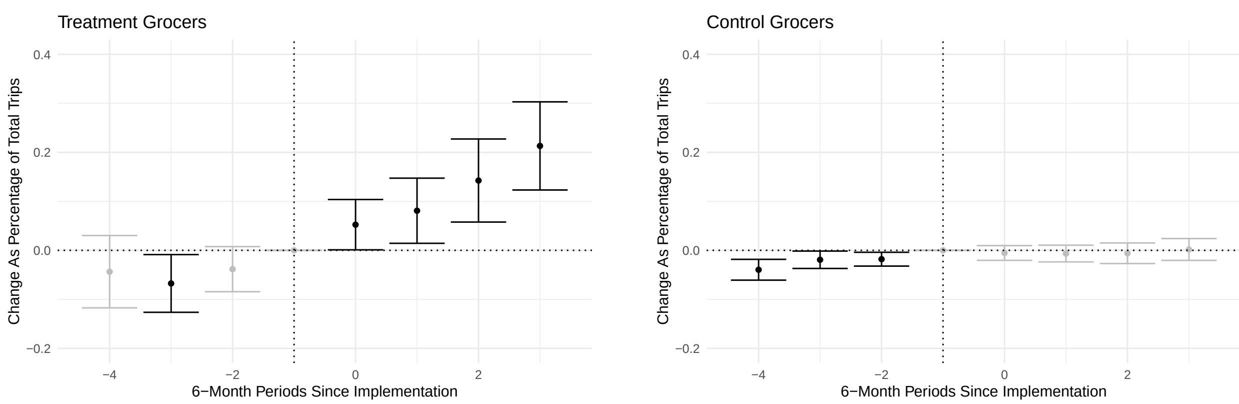
Reply: I use a Poisson specification following Cohn et al. (2022), who show it is preferred for difference-in-differences designs with count-like outcomes bounded below by zero. OLS in levels estimates absolute changes, not percentage changes, and OLS in logs is not feasible because the outcome contains many zeros.

An appropriately designed linear model does yield similar estimates. Figures A.1a and A.1b run separate OLS difference-in-differences for the focal and control grocers. Subtracting the percentage estimates (using pre-period means) yields approximately the same result as the Poisson triple-difference. Poisson is more convenient because it directly estimates percentage effects and scales to multiple events without the two-step conversion.

Figure A.1: OLS difference-in-differences estimates for credit card acceptance effects

(a) Focal grocer (accepted credit cards)

(b) Control grocers (always accepted)



Notes: OLS-style event study coefficients for credit card users relative to other consumers. The left panel shows results for the focal grocer that began accepting credit cards, and the right panel shows results for control grocers that always accepted credit cards. Subtracting these percentage estimates yields an effect comparable to the Poisson triple-difference estimate in the main text.

Why isn't the credit card share at the large grocery equal to zero before it began accepting credits cards (Figure 4a)? If there is measurement error in the data, how does this affect the author's estimation procedure?

Reply: I infer acceptance from large shifts in credit card payment shares and classify a merchant as accepting credit cards only if more than 5% of its sales are paid by credit.

The nonzero credit share before the acceptance change reflects misreporting: consumers sometimes confuse signature debit cards with credit cards when responding to surveys (Rysman et al. 2025). As shown in Appendix Table OA.2, self-reported payment shares in HomeScan align well with aggregate payment volumes, so this individual-level noise does not bias the estimation of the payment patterns the model targets. I have also added a direct cross-dataset check of the multi-homing statistics in Appendix Table OA.5: the usage-based Homescan multi-homing rate (56.7%) matches the ownership-based rate from the Diary of Consumer Payment Choice survey (57.7%) to within one percentage point over 2017–2019, so the Homescan single-homing share is not driven by usage-based misclassification.

> The first sentence of Section III.B.2, that "Credit card users shop more at the grocer because consumers value the flexibility of paying with credit — not because of rewards" is imprecise and not entirely clear to me. The author showed that consumers are sensitive to rewards, which are thus likely a large motivator of credit card usage. Thus, the flexibility of paying with credit is likely valued by consumers because credit card usage entails rewards. The author also states that the flexibility of paying with credit may be valued by consumers because consumers like to make large purchases with credit, but this begs the question. There must be some reason why consumers like to make large purchases with credit (perhaps because they dislike using credit, as the author argues, but the rewards accumulated on large purchases are especially high). The conclusion that I draw from Figure 5 is that credit cards are highly substitutable with each other but weakly substitutable with other payment methods.
>
> The author concludes from Figure 5(a) that higher Discover rewards do not affect consumer choices about which stores to visit. I have two comments on this conclusion:
>
> The author focuses only on grocery stores. Perhaps consumers' food shopping is not impacted by rewards given that food is a necessity, but other areas of spending would be affected. Can the author repeat the analysis for other categories?
>
> I am not even sure that the conclusion is correct. It looks like there are upticks in the black curve during the periods in which Discover offers higher rewards for groceries. Then again, the dashed line also seems to increase. It would be helpful for the author to estimate the differential impact of higher Discover rewards on Discover users versus non-Discover users using a fixed effects regression; this would allow the author to conduct inference on the impact of higher rewards on the number of trips.

Reply: The Discover evidence shows that rewards affect payment method choice but not store choice. Consumers switch to Discover to earn rewards but do not shift spending toward grocery stores. You are right that this does not "rule in" a particular mechanism for why credit card acceptance increases sales.

I have added a fixed effects regression in Online Appendix OD.2 with household and time fixed effects, clustering standard errors at the household level. The key coefficient on "Discover HH $\times$ Grocery Reward Month" is small and economically negligible for the grocery share of total household spending (0.002, SE 0.001), but the Discover share of grocery spending increases by 7.6 pp (SE 0.003). Consumers shift payment methods *within* grocery stores toward Discover but do not change *where* they shop.

It is not possible to study other categories. A credible design requires cases where Discover pays rewards at some stores but not at adequate substitutes. Grocery stores are a target of the rewards program, and Ellickson et al. (2020) show that grocers and warehouse stores are substitutes for food purchases. No other category offers this structure.

> In Figure 6, a line labelled "OptBlue" appears, but "OptBlue" is never discussed in the main text (although it is summarized in a figure note). I fail to see the point of including this second line as it does not influence the author's argument in Section III.C.1.

Reply: The OptBlue line serves two important roles. First, it explains why the model treats AmEx acceptance as comparable to Visa acceptance, despite the historical impression that AmEx is accepted less often. The OptBlue program cut AmEx merchant fees for small businesses starting in 2015, and Figure 5b shows that the Visa-AmEx acceptance gap closed shortly after. By the 2019 calibration year, nearly all merchants accept either all credit cards or none. Second, I use the OptBlue episode as an out-of-sample validation of the model's merchant fee sensitivity estimates in Section V.C.2. The model's predicted acceptance response to the fee cut matches the observed convergence. I have clarified this dual purpose in the text.

> What does the author mean "Since every Visa/MC merchant also accepts AmEx"? From experience, many merchants refuse AmEx but accept Visa and MC. Also, doesn't this statement clash with the author's preceding statement that merchant adoption strategies tend to follow a hierarchical pattern? Perhaps the author meant to say that "every AmEx merchant also accepts Visa/MC"? Even if so, it seems hard to believe that there is not a miniscule share of merchants that accept AmEx but not Visa/MC.

Reply: You are correct that historically many merchants refused AmEx but accepted Visa and MC. However, as Figure 5b shows, this gap has largely closed between 2016 and 2019. The number of merchants accepting AmEx now approximately equals the number accepting Visa. I have revised the text to clarify that the statement refers to the current state of the market, not historical patterns, and to note that this represents a change from the past.

> Figure 6 does not directly tell us about the extent of merchant multihoming. What we would need to assess this extent are additional lines showing the number of merchants on various combinations of the networks (Visa + MC, Visa + Amex, etc.).

Reply: I agree that detailed data on every possible acceptance subset would be ideal, and that Figure 5 alone is not sufficient to fully assess merchant multihoming.

However, unlike platform markets such as food delivery, where merchants routinely accept some platforms but not others, the card payment industry does not exhibit this pattern. Merchants that accept credit cards overwhelmingly accept all major networks, and very few accept Visa but not MC or vice versa. Precisely because combinatorial acceptance is not a feature of this industry, no data source systematically records the acceptance subsets of individual merchants.

The best available evidence comes from Yelp reviews in Online Appendix OD.3. Using around 1,500 reviews that mention at least two payment methods, I find that merchants progressively add payment methods (cash $\rightarrow$ debit $\rightarrow$ Visa/MC $\rightarrow$ AmEx) in a hierarchical pattern rather than specializing in particular networks. Combined with Figure 5b showing that the number of AmEx-accepting merchants now approximately equals the number accepting Visa, the data support a world with limited heterogeneity in acceptance bundles.

> It does not immediately follow that large effects of card adoption on sales mean that merchants should be insensitive to fees. As the author notes in Section III.B.3, whether the sales increase from card adoption makes adoption worthwhile to a merchant depends on the merchant's margins (as well as the merchant's fixed costs of adoption). If, e.g., there was a large mass of merchants with margins around 20% and no fixed costs of adoption, then small changes in fees would change the adoption decision of many merchants. The elasticity of adoption with respect to fees similarly depends on the distribution of fixed costs of card adoption.

Reply: I agree. The sales increase from card acceptance is not sufficient to pin down the merchant fee elasticity. The elasticity depends on the distribution of merchant margins and fixed costs of adoption, which I do not directly observe. I have revised the text to be more transparent about this limitation and to clarify that the merchant fee sensitivity is inferred from the model's equilibrium conditions and validated against the OptBlue episode, rather than being directly estimated from the reduced form evidence. In the model, I assume all merchants have the same profit margin through one CES elasticity σ.

> Is anything lost by assuming that card users never prefer to pay with cash? As the author notes, the consumer's random selection to use either the primary or secondary card could arise from a model with preference shocks at the level of a transaction and payment method. It would seem natural for there to be some shock for using cash as well. Perhaps the author may overstate the benefits to consumers of payment cards by ruling out the possibility that sometimes consumers would prefer to use cash?

Reply: Yes, the model overstates the welfare gap between pure card users and pure cash users by polarizing consumers into extreme types. A richer model with transaction-level cash preference shocks would place consumers on a continuum. In the HomeScan data, roughly 25% of purchases are made with cash regardless of how many cards consumers carry, consistent with such shocks.

The aggregate welfare results are robust to this concern. Several offsetting forces preserve their direction. Credit aversion per card holder is overestimated: card holders use cards less often than modeled, so they bear credit aversion costs less frequently. But the number of card holders is underestimated, because the richer model needs more card holders to match observed spending shares. The rewards loss per card holder from a fee cap is similarly overstated, but this overstatement is spread across a larger number of card holders. These forces partially offset, so the main conclusions (that fee caps benefit consumers on net, with progressive distributional effects) are directionally robust.

> Page 18 reads "Second, consistent with the evidence in Section III.B.1, rewards do not affect relative consumption choices across merchants." But it seems that Section III.B.1 ("The Effects of Accepting Credit Cards") is about how credit card acceptance by a single large merchant affected sales at the merchant, not about rewards or a comparison of consumption across merchants. Perhaps the author meant to refer to Section III.B.2.

Reply: I have corrected the cross-reference to refer to the Section on the Discover evidence.

> Section V.C.1 states that "my estimated retail margin of 15.6 percent is also similar to the aggregate markups of 15–20% used in macro studies of misallocation." However, Section III.B.3 reads "credit card acceptance is profitable as long as margins exceed 20%. Census statistics indicate gross margins in the grocery industry were around 27% during this period." The author should fix the consistency between these parts of the paper, as the earlier argument that merchant should be insensitive to fees seems to rely on margins exceeding 20% (although, as discussed, this argument is flawed). Also, the author should also be consistent in writing out "percent" or using the "%" sign.

Reply: I have removed the back-of-the-envelope margin calculation from the reduced-form section. You are correct that the earlier argument was both inconsistent and relied on strong assumptions on the distribution of merchant margins. The reduced-form evidence is not sufficient to pin down the fee elasticity, so I have removed the attempt to do so from that part of the paper.

In the estimation section, I now clarify that the margin is estimated structurally: the model recovers a retail margin of 17.8 (SE 5.0), which is close to the 13–17% margin range implied by the 15–20% markups cited in macro studies of misallocation. I also clarify the identification logic: because credit and debit are segmented, consumers who would use credit instead pay cash when credit is not accepted, and debit transactions are unaffected. The marginal merchant therefore weighs credit fees against sales that would otherwise be lost to cash, and it is this credit-versus-cash gap that pins down the retail margin. I have also standardized notation to use % consistently throughout.

> The claim that "Capping fees at the cost of cash also simulates the effects of merchants freely surcharging consumers for the cost of card acceptance" in Section VI.A requires greater explanation.

Reply: As shown by Zenger (2011), capping merchant fees at the cost of cash is theoretically equivalent to allowing merchants to freely surcharge consumers for the cost of card acceptance. However, the revised draft no longer emphasizes this equivalence in the main text. The primary counterfactual now caps credit fees at 120 bp, an intermediate cap that keeps the analysis close to observed fee levels. Because this cap exceeds the cost of cash, the surcharging equivalence does not apply directly. For completeness, Online Appendix OG.4 compares the intermediate cap to the cost-of-cash benchmark, where the surcharging equivalence holds: perfect surcharging makes the allocation of fees between merchants and consumers irrelevant, so the two equilibria coincide only at that specific cap level.

> A more appropriate name for Section VI.C would be "Reducing Competition Between Private Networks."

Reply: I have revised the section title to "Welfare Effects of Reducing Network Competition," which more precisely conveys the economic content.

Remaining Follow-up Points from Original Report

The following points were assessed as satisfactory, but I address some of the ones that I think require further clarification.

> For information on payment method choice, the article uses the Atlanta Federal Reserve's Diary of Consumer Payment Choice and the Survey of Consumer Payment Choice. How were the data combined? Do they include the same consumers, or did the author pool together data from two distinct sources?

Reply: There is complete overlap between the diary and survey respondents within each wave: the same individuals complete both the diary and the survey in a given year. I have clarified this in Section II.B by stating that "the diary and survey share a common panel of respondents" rather than the ambiguous "common set of identifiers."

> There is little said in the data section about the availability of data on merchants. It is unclear how the author can credibly estimate a model of merchant card acceptance without direct data on merchant acceptance.

Reply: The Diary of Consumer Payment Choice measures merchant acceptance transaction by transaction. For each purchase, the respondent records which payment methods the merchant would have accepted. Because the diary asks respondents to log every transaction over a three-day window in real time, this is not a retrospective recall question but a record made at the point of sale. The resulting acceptance rates are sales-weighted averages across the merchants each consumer visits. The model requires exactly this object: the probability that a randomly drawn transaction occurs at a merchant accepting a given network. A new footnote in Section II.B acknowledges that the measure is imperfect. It excludes merchants the consumer did not visit and relies on the consumer correctly observing acceptance. As a cross-check, Online Appendix Table OC.3 compares DCPC acceptance rates to Yelp business attributes by sector. The two sources agree closely in aggregate (95.8% vs. 96.3%), which suggests that consumer reporting in the diary is reasonably accurate. Transaction-level merchant acceptance data are rare, and the diary's real-time design together with the Yelp validation mitigate the most serious recall concerns.

> The author has revised the procedure for determining first and second cards as detailed by Appendix A.3.1 in light of the comment. However, this appendix is missing details. Several parts are incomplete (the first sentence of the appendix is incomplete; the final sentence of the "Cash-only consumers" section is incomplete; the section describes various cut-offs without mentioning their numerical values).

Reply: I have revised Appendix A to fix the incomplete sentences and include all numerical cutoff values used in the classification procedure.

> The author states "I hold fixed merchant actions because my difference-in-difference estimates remove any equilibrium effects of merchant actions." Some additional explanation would be helpful.

Reply: The intuition is as follows. The difference-in-differences design compares debit card volumes at regulated versus exempt issuers. Both groups of issuers' cardholders face the same merchant acceptance environment. Any changes in merchant behavior after Durbin affect all consumers equally and are differenced out by the control group. The model-simulated Durbin moment therefore holds merchant acceptance fixed when predicting how a reduction in debit rewards would change debit card volumes, because the empirical target has already purged this channel. I have expanded this explanation in Section V.A.1.

References

Cohn, Jonathan B., Zack Liu, and Wardlaw. 2022. "Count (and Count-like) Data in Finance." *Journal of Financial Economics* 146, no. 2 (November): 529–551.

Ellickson, Paul B., Paul L.E. Grieco, and Oleksii Khvastunov. 2020. "Measuring Competition in Spatial Retail." *The RAND Journal of Economics* 51, no. 1 (March): 189–232. Accessed October 9, 2024.

Rysman, Marc, Shuang Wang, and Krzysztof Wozniak. 2025. "Understanding Preferences for Payment Cards Using Household Scanner Data." *Working Paper.*

Teh, Tat-How, Chunchun Liu, Julian Wright, and Junjie Zhou. 2023. "Multihoming and Oligopolistic Platform Competition." *American Economic Journal: Microeconomics* 15 (4): 68–113.

Zenger, Hans. 2011. "Perfect Surcharging and the Tourist Test Interchange Fee." *Journal of Banking & Finance* 35, no. 10 (October): 2544–2546.